

Repairs implemented. The remaining mathematics mapped.

A reproducible Lean project and a book-wide formalization backlog for Statistical Consequences of Fat Tails, third edition.

DELIVERED	CHECKED SCOPE
16 replacement entry points	Shared, importable implementations for Proof01 through Proof16.
50 theorem declarations + 1 instance	Aggregate build and complete printed axiom closure pass. Sixteen aliases expose the original file organization.
158 obligation families	Source anchors, hypotheses, dependency groups and priorities across 30 numbered and 9 lettered chapters.
New probability bridges	Self-convolution subexponentiality, independent-sum laws and conditional stable-law identification.

The previous audit ZIP already contained implemented repairs. This handoff retains them, improves the probability interfaces, adds small reusable lemmas, and makes reproduction and book coverage easier to inspect.

The complete book is not formalized. Some source statements need correction before proof search. This package does not establish stable-law existence for all parameters, acceptance into Mathlib, or readiness of a separate quantum/classical application.

Deliverables: Taleb_Lean_Implementation_Handoff.zip and this report. The ZIP contains all source, documentation, JSON indexes, verification scripts, and compiler evidence.

Source: arXiv:2001.10488v4, 17 September 2025; 523 PDF pages. Formalization inventory date: 18 September 2026.

How to reproduce and use the handoff

Extract the ZIP, enter Taleb_Lean_Repairs, and run these commands with elan/Lean installed. The first command downloads pinned dependencies and precompiled Mathlib artifacts.

```
lake exe cache get
lake build
python3 scripts/verify.py
```

Lean version: 4.24.0. Mathlib commit:

f897ebcf72cd16f89ab4577d0c826cd14afaafc7

Keep lean-toolchain and lake-manifest.json. The included project version applies even when the user has a newer default Lean installation. No dependency upgrade is needed to reproduce this handoff.

Verification result

PASS: 50 theorems, 1 instance, 16 aliases. The script checks dependency commits, an aggregate Lake build and freshly evaluated axiom queries for all 67 proof declarations. Their dependencies are restricted to propext, Classical.choice and Quot.sound. No sorryAx is accepted.

Fifty theorems does not mean fifty new book proofs: the count includes diagnostic counterexamples, simpler variants and foundation lemmas. Aliases add no independent mathematical content. The verification JSON records hashes of every Lean source file.

evidence/current/build.log, axioms.log and verification.json are fresh handoff evidence. The older evidence directory intentionally retains original-file failures: the submitted archive had eight passing and eight failing files on this baseline.

Import the complete project

```
import AuditRepairs
```

```
#check Taleb.Proof01.repaired
#check Taleb.Proof08.repaired
#check Taleb.Proof16.repaired
```

The entry points share definitions, preventing the duplicate global declarations in the original standalone files. False or underspecified original statements have deliberately changed signatures; see docs/REPLACEMENT_MAP.md before updating client code.

Original input archive SHA-256

b0132c09a8835107abaac916578e5e61a703
b5a09a800c2e874c24d15d05b47c

Replacement map for all sixteen files

The map gives the principal replacement. Supporting original-scope identities remain in the shared modules. The exact declaration names and import paths are in the ZIP.

FILE	REPLACEMENT	CHANGE / LIMIT
01	Log is slowly varying	Retain sound argument; simpler algebra.
02	Nonzero constant is slowly varying	Remove failing no-op tactic.
03	Nonzero limiting constant implies slow variation	Correct limit composition.
04	Product of regular-varying functions	Retain the ratio argument.
05	Real power closure	Fix continuity and exponent order; eventual positivity.
06	RV iff power-normalized SV	Shorter shared proof; unnecessary sign hypothesis removed.
07	Finite Pareto tail exponent	Explicit positive-tail finite-limit interface.
08	Subexponential convolution tail exponent	Now uses a probability law and its self-convolution.
09	Convexity of $c^{(-\alpha)}$	Qualify <code>Set.univ</code> and prove the correct statement.
10	Frechet maximum-scaling identity	Global support-aware formula; $n > 0$.
11	Gumbel maximum-scaling identity	Correct original argument retained.
12	Gaussian specialization	Correct parameter order; repaired S1 expression.
13	Gaussian exponential scaling	Sound algebra, simplified.
14	Cauchy specialization	Correct parameter order and nonnegative scale.
15	Cauchy exponential scaling	Sound algebra, simplified.
16	Stable convolution-power law	Conditional measure equality via corrected S1 function.

Proof09, Proof12 and Proof14 have checked counterexamples to the original statements. Their intended corrected statements are proved. The `ImplicitSetDiagnostic` module intentionally retains `autoImplicit` to reproduce the old error; production repair modules disable it.

What the added code actually proves

Eleven small foundation results

Foundations.lean adds: the zero-index equivalence; pure-power regular variation; invariance under eventual equality and nonzero scaling; slow-variation product and power closure; uniqueness and liminf agreement for finite tail exponents; the analytic Pareto power-transform exponent; and the corrected shifted-gamma excess identity and its positive sign.

A probability-level subexponential interface

IsSubexponential now requires a probability measure on the nonnegative reals, an eventually positive survival function, and a limit of its self-convolution tail divided by its own tail equal to 2. The checked theorem preserves an existing finite tail exponent under that self-convolution. It does not prove that every regularly varying distribution is subexponential; that is a separate backlog family.

Convolution and stable-law identification

The package defines zero-fold convolution as the point mass at zero, proves convolution powers are probability measures, proves their characteristic functions are powers, and reuses Mathlib to identify the resulting stable law when both required characteristic-function premises are supplied. An independent-sum theorem connects random-variable laws to convolution.

S1 normalization and specializations

The corrected piecewise expression includes the skewed $\alpha=1$ logarithmic branch. Its Gaussian and centered Cauchy specializations and its value 1 at $t=0$ are checked. Stable scale and Gaussian standard deviation are different: $\exp(-\text{scale}^2 t^2)$ has variance $2 \cdot \text{scale}^2$.

Remaining semantic obligations

Stable-law existence is still open in this project. Frechet/Gumbel expressions still need measure construction and iid maximum-law bridges. General infinite or undefined tail indices still need an extended-real design. The safe finite interface is used in the new results; legacy definitions remain only for comparison and diagnostics.

The gamma mixture formula is repaired algebraically. Its inverse-moment integral and finiteness threshold are recorded as outstanding probability work. This distinction prevents a rational identity from being reported as a proved mixture theorem.

Mathlib first: reuse before extending

These are facts about the pinned source snapshot. No absence claim or contribution-novelty claim is made about current Mathlib master. Check upstream again before proposing a contribution.

AREA	ALREADY AVAILABLE	NEXT CONTRIBUTION
Characteristic functions	Convolution product, uniqueness of finite measures, independent-sum law	Stable measure existence and parameter API; reuse the bridge machinery.
Gaussian distributions	Gaussian characteristic function and Gaussian convolution closure	Specializations and applications, not duplicate basic closure proofs.
Tails and integrals	Layer-cake integrals, real powers, gamma/beta integrals	Pareto moments, threshold excess and explicit divergence criteria.
Pareto / CDF	Pareto probability measure, density/CDF bridge, CDF uniqueness	Exact survival/moment corollaries, iid extrema and EVT laws.
LLN / convexity	Strong laws and integral Jensen	Estimator corollaries and moment convexity with all hypotheses.
Regular variation	Asymptotic equivalence, filters and power continuity	Coherent positive/measurable API; uniform convergence, Potter and Karamata.

Existing declarations directly used or identified

charFun_conv; Measure.ext_of_charFun; IndepFun.map_add_eq_map_conv_map0;
charFun_gaussianReal; gaussianReal_conv_gaussianReal; isProbabilityMeasure_paretoMeasure;
Integrable.integral_eq_integral_meas_lt; ConvexOn.map_integral_le; strong_law_ae_real. Exact source modules are in docs/MATHLIB_AND_WORK_ORDER.md.

Large dependencies

Generalized CLT, stable domains of attraction, Karamata/EVT, statistical estimator asymptotics, differential-entropy optimization and stochastic calculus should be treated as substantial theorem families. A file containing a plausible formula is not a completed contribution in any of these areas.

Work order and the near-term objective

STEP	CONCRETE TARGET	DEPENDENCIES / EXIT
1	Reproduce the repaired package	Pinned build and all 67 proof-declaration axiom queries pass.
2	Exact Pareto tail and moment API	Use existing paretoMeasure; prove integrals and divergent cases.
3	iid maxima/minima and valid EVT measures	Use independence and CDF/Stieltjes infrastructure; connect formula identities.
4	RV and subexponential infrastructure	Agree definitions; add uniform convergence, Potter bounds, Karamata and closure.
5	Stable laws and limit theorems	Construct laws; prove domains of attraction and sample-average results.
6	Book applications	Kappa, Gini, mixtures, quantiles, shadow moments and selected pricing results.

Good first backlog slices

T005: threshold tail integrals; T032: power-transform pushforward; T060: iid maximum/minimum laws; T068: exact Pareto MLE; T118: Pareto moment convexity; T124: shifted-gamma inverse moments. Pick one small slice after examining its prerequisites, instead of tying all six to one deadline.

A realistic weekend result

Reproduce this handoff, choose one contained reusable contribution, and settle its mathematical statement. Full-book formalization and Mathlib review are separate milestones; upstream acceptance is not controlled by the local build.

The Monday software objective

This Lean project uses noncomputable real and measure-theoretic objects. It is not executable quantum/classical application software. Work on that application needs its repository, runnable baseline, input/output examples and the exact paper claim to demonstrate. A local demo should have its own acceptance test and clearly identify which mathematical contract, if any, is connected to its computation.

The handoff is designed to be passed directly to your friend for reproduction and further implementation.

Source statements to repair before formalization

The following checks are targeted findings from the inventory pass. They extend the initial audit but do not constitute a complete correctness audit of the book. Except for explicitly named Lean lemmas, these are mathematical checks recorded for the next implementation stage. Printed page numbers below need +14 for the PDF viewer.

G01. Slow variation is not convergence to a constant

Sections 2.2.1 and 21.2. The function $\log(x)$ is slowly varying and diverges.

`AuditRV.isSlowlyVarying_log` is already checked. Eventual constant-tail approximations require an additional quantitative or asymptotic assumption.

G02. Exponential is a boundary, not a member of the heavy-tail subexponential class

Pages 91-93. If X, Y are independent exponentials of rate $r > 0$, $P(X+Y > x)/P(X > x) = 1 + r \cdot x$ for $x \geq 0$. It diverges instead of tending to 2. Use the convolution definition in

`AuditProbability.IsSubexponential`. Gamma/Laplace exponential tails do not become heavy-tail subexponential merely because they are mentioned alongside a boundary case.

G03. The moment boundary of general regular variation is undecided by alpha alone

Pages 91-98. For nonnegative moment order q and survival $S(x) = x^{(-\alpha)}L(x)$, $q < \alpha$ gives the usual finite-moment result, and $q > \alpha$ divergence. At $q = \alpha$ the deciding integral is proportional to $\int L(x)/x \, dx$. For example, a normalized eventual survival proportional to $x^{(-1)}/(\log x)^2$ has index 1 and finite mean. Exact Pareto statements cannot be silently generalized at the boundary.

G04. Product tails need assumptions

Page 100, Principle 5.2. Multiplication by the thin-tailed variable $Y=0$ yields zero, not a heavy tail. The useful replacement is a Breiman-type theorem for independent nonnegative variables with a finite $(\alpha + \epsilon)$ moment of Y and a positive α moment. Dependence and cancellation need separate treatment.

G05. Stable absolute moment fails the Gaussian endpoint check

Page 152, equation (7.9), visually checked. With $\alpha=2$, $\beta=0$ and location zero, the displayed formula gives $\text{scale}/\sqrt{\pi}$. The chapter's characteristic function is $\exp(-\text{scale}^2 t^2)$, corresponding to Gaussian variance $2 \cdot \text{scale}^2$ and $E|X| = 2 \cdot \text{scale}/\sqrt{\pi}$. Resolve this factor of two before formalizing the general stable moment derivation. The delivered S1 specializations fix the parameter-order errors in the original Lean files, but do not prove the corrected general absolute-moment integral.

G06. A hidden finite moment cannot have $p > \alpha$ for an exact Pareto tail

Pages 177-178, Proposition 9.1 and equation (9.12), visually checked. The hidden moment is defined as $\int_0^\infty x^p f(x) \, dx$. For Pareto density $\alpha \cdot L^\alpha \cdot x^{(-\alpha-1)}$, this equals $\alpha \cdot L^\alpha \cdot K^{(p-\alpha)}/(\alpha-p)$ only when $p < \alpha$. It diverges for $p \geq \alpha$. The printed $p > \alpha$ domain cannot give a real-valued finite hidden-moment density. Re-derive the distribution in the valid finite-moment regime and distinguish exact Pareto from approximation.

G07. Full support does not force a Laplace log-growth law

Page 184, Theorem 1, visually checked. The one-sided result $\log(X/L) \sim \text{Exponential}(\alpha)$ follows from exact Pareto X . Allowing the log variable to range over all reals does not determine its negative side, symmetry, or mixing weights. An independent bilateral/symmetry model is required for a Laplace conclusion.

G08. Shortfall ratios have sign and normalization errors

Page 200, Definition 10.1, and page 228, Definition 11.6, visually checked. For a Pareto law with $\alpha > 1$, $E[X \text{ given } X > K]/K = \alpha/(\alpha-1) > 1$ and $E[X-K \text{ given } X > K]/K = 1/(\alpha-1)$. A second division by K changes the limit. The printed $\alpha/(1-\alpha)$ is negative and cannot be the ratio of positive tail payoffs. The adjusted payoff-equivalent factor may exceed one; it is then not a probability.

G09. The sigmoid martingale statement is false as written

Page 268, Proposition 14.1, visually checked. Let X_t be constantly 1. Both X_t and $\text{logistic}(X_t)$ are martingales, but $\text{logistic}(1)$ is not $1/2$, refuting the literal only-if statement. Even for nondegenerate Brownian motion started at zero, symmetry of the unconditional mean does not prove the sigmoid process is a martingale. For $dX = b(X)dt + \sigma(X)dW$, the drift of $f(X)$ is $f'(X)b(X) + (1/2)f''(X)\sigma(X)^2$. A local drift condition over the reachable states, plus integrability, is the right target.

G10. Gini ratios require joint fluctuation analysis

Pages 283-285 and 295, Theorem 2 and equation (15.42). The numerator and denominator share observations. At stable order, denominator fluctuations cannot be dropped as if they were smaller. The appendix itself introduces $X_i^*(2F(X_i)-1-g)$. Verify its tail constants and scaling before accepting the displayed limit parameter. This is a source-review gate, not a completed counterexample or repaired Gini theorem.

G11. Strict bias and monotonicity need stronger hypotheses

Page 305, Theorem 5. If every observation lies above the fixed threshold h , the empirical and population threshold shares both equal 1; strict downward bias fails. Also, superadditivity of $n \cdot a_n$ alone does not imply that a_n increases: $b_n = \text{floor}(n/2)$ is superadditive, while $b_{2/2} = 1/2 > b_{3/3} = 1/3$. Consistency from an appropriate LLN is a separate, plausible claim and should be proved independently.

G12. Shifted-lognormal mixture mean has an invalid generalization

Page 385, equations (21.11)-(21.13), visually checked and carried forward from the first audit. Set $b=2$, $\alpha_0=3$, $\log\text{-variance}=1$ and $\text{scale}=1$. Every conditional Pareto mean for $\alpha=2+Z$, $Z>0$, lies in $(1,2)$. The printed closed form gives $1+e>2$. The correct general expression is $\text{scale} \cdot (1 + E[1/(b-1+Z)])$. At $b=1$ and $E[Z]=\alpha_0-1$ it simplifies to $\text{scale} \cdot (1 + \exp(\log\text{-variance})/(\alpha_0-1))$. Termwise integration of the proposed lognormal series needs an independent convergence argument.

G13. Shifted-gamma correction needs scale and a finite inverse moment

Page 386, equations (21.14)-(21.16), visually checked and carried forward. Set $m=\alpha_0-1$ and let $\alpha-1$ have gamma shape m^2/s^2 and scale s^2/m . For $m^2 > s^2$, the mixture mean is $\text{scale} \cdot (1 + m/(m^2 - s^2))$; its excess over the fixed-index mean is $\text{scale} \cdot s^2/(m \cdot (m^2 - s^2))$. The displayed equation repeats the primed expectation and omits scale. If $m^2 \leq s^2$, the inverse moment diverges despite $\alpha > 1$ almost surely. `AuditMoments.gamma_mixture_excess` and `gamma_mixture_excess_pos` check the repaired algebra and sign. The gamma integral and its divergence threshold remain backlog items.

G14. The minimum-p-value expectation formula has the wrong sign and integrand

Page 395 after Proposition 22.3, visually checked. The displayed integral is negative for a positive density; at $m=1$ it is -1 . For m independent $[0,1]$ values with CDF F , $E[\min] = \int_0^1 (1-F(p))^m dp$, equivalently $\int_0^1 p^m f(p) (1-F(p))^{m-1} dp$ when a density exists. Identical marginal distributions alone do not give the product survival law.

G15. A forward mean does not identify a probability measure

Page 426, Lemma 25.3 and equation (25.13), visually checked. The point mass at 1 and the equal mixture of point masses at 0 and 2 have the same first moment and different laws. Equality of forward expectations cannot imply a Radon-Nikodym derivative of 1. The payoff identity is $(x-K)_+ - (K-x)_+ = x-K$. A full arbitrage-free option curve across strikes can determine a terminal marginal; a single mean cannot, and neither identifies an entire path law.

G16. Absolute-moment constraint has a sign error

Page 478, Section 30.4.2, visually checked. Under the chapter's $K < 0$ setting, the left tail is negative, so its contribution to $E|X|$ is $-\epsilon \nu_-$. The printed constraint uses $+\epsilon \nu_-$. Correct this sign before solving for the truncated-Laplace rate or proving maximum entropy.

How to read the remaining-mathematics inventory

The 158 rows below are formalization families. Each may expand into definitions, integrability lemmas, convergence machinery, a main theorem and corollaries. They are not a promise of 158 independent missing theorems or an exhaustive atomic extraction of every implicit argument.

STATUS	MEANING	#
missing	Implement after prerequisites.	95
source-check	Resolve the mathematical statement first.	33
partial	Some delivered ingredients exist; family incomplete.	8
reuse	Instantiate or connect existing Mathlib facts.	4
model-needed	Specify a mathematical or software model first.	15
empirical	Reproduce data/calculations; not a proof-only task.	3

Priorities: P0 = statement repair; P1 = comparatively contained reusable family; P2 = substantial dependency family; P3 = advanced application, model or data work. A source-check status always requires mathematical review, regardless of priority. These are dependency priorities, not day estimates.

Coverage and locator evidence

All 39 chapter/lettered units have coverage entries. Chapter 1 is introductory exposition. The source indexes contain 353 PDF bookmarks, 315 numbered-equation locator occurrences and 71 statement-heading candidates. Equation and heading indexes can include repeats or prose references; they are navigational evidence, not theorem counts.

Every source body page through printed page 482 was included in the locator scan. Unnumbered mathematics is represented through the curated families and needs further atomization when implemented. Source-specific hypotheses and corrections are included in each row. The scan and curated inventory have reproducible Python scripts in the ZIP.

The printed page ranges below are source anchors, not assurances that every needed dependency is on that page. The JSON gives a PDF anchor page as well. PDF bookmarks can precede the exact formula; consult the rendered source before choosing a precise theorem statement.

Dependency groups: RV regular variation; TAIL survival integrals; MOM moments; CF characteristic functions; CDF extrema/distribution functions; QUANT quantiles; LLN laws of large numbers; CLT limit laws; EVT extreme values; SUBEXP convolution tails; GAMMA special functions; DENS densities; CALC calculus/convexity; KAPPA metric; EST estimators; GINI inequality; MULTI multivariate laws; ENT entropy; RUIN survival/barriers; SDE stochastic processes; FIN pricing; APP model/data applications.

The outstanding formalization families

Chapter 2

T001 Regular variation API

P1 | partial | RV | Printed pp. 9-10 | Source 2.2.1

Separate ratio limits from eventual positivity and measurability; connect the positive measurable convention to the repaired predicates.

Hypotheses / gap: Measurable functions; eventual positive values; positive scaling argument.

T002 Karamata representation

P2 | missing | RV | Printed pp. 9;379-382 | Source 2.2.1;21.1-21.3

Prove the integral representation of measurable slowly varying functions and its converse.

Hypotheses / gap: Positive measurable L ; locally integrable $\epsilon(t)/t$; ϵ tends to zero; η has finite limit.

T003 Weak and strong laws

P1 | reuse | LLN | Printed pp. 10;137-139 | Source 2.2.2;7.1

Instantiate existing laws for integrable iid real variables and their sample averages.

Hypotheses / gap: Identical laws; appropriate independence; integrability; n positive.

T004 Central limit theorem

P2 | missing | CLT | Printed pp. 10;147-148 | Source 2.2.3;7.5

Encode normalized sums and convergence in distribution to a Gaussian.

Hypotheses / gap: Independence; finite nonzero variance; distinguish iid and triangular-array versions.

T005 Tail integral and excess identities

P1 | missing | TAIL | Printed pp. 10;18;259 | Source 2.2.4;2.10;13.6

Prove $E[X \mathbb{1}_{(X>K)}] = K P(X>K) + \int_K^\infty P(X>x) dx$; derive excess and conditional means.

Hypotheses / gap: Nonnegative X and K ; Tonelli for extended integrals; finite first moment for real-valued identities; positive tail for conditioning.

T006 Kappa metric domain

P1 | missing | KAPPA | Printed pp. 11;156-159 | Source 2.2.6;8.2

Define kappa only where $n>n_0>0$, $M(n_0)>0$, and $\log(M(n)/M(n_0))$ is nonzero.

Hypotheses / gap: Finite absolute first moment; nondegenerate dispersion; no unrestricted $[0,1]$ range claim.

T007 Stable law existence

P2 | missing | CF | Printed pp. 12-13;139-140 | Source 2.2.9;7.2.1

Construct the probability measure with the repaired S1 characteristic function.

Hypotheses / gap: $0<\alpha\leq 2$; $-1\leq\beta\leq 1$; $\text{scale}\geq 0$; include $\alpha=1$ logarithmic branch and zero-scale Dirac case.

T008 Subexponential law API

P2 | partial | SUBEXP | Printed pp. 13;91-93 | Source 2.2.12;5.1

Extend the new nonnegative self-convolution definition to the standard equivalent n -fold tail characterizations.

Hypotheses / gap: Probability law supported on nonnegative reals; unbounded right support; fixed positive integer n .

T009 Quantiles and expected shortfall

P1 | missing | QUANT | Printed pp. 16 | Source 2.2.19

Use generalized inverses and state the atom-safe expected-shortfall formula.

Hypotheses / gap: Probability level in $(0,1)$; integrability; distinguish strict/non-strict tails and atoms.

T010 Maximum divided by sum

P2 | missing | LLN | Printed pp. 16-17;200 | Source 2.2.21;10.2.6

Show maximum X_i^p divided by sum X_i^p tends almost surely to zero.

Hypotheses / gap: Nonnegative iid observations; finite strictly positive p th moment; define initial zero denominators.

T011 Maximum domain of attraction

P1 | source-check | EVT | Printed pp. 17;171-177 | Source 2.2.22;9.1

Define normalized maxima convergence; distinguish it from convergence to the support endpoint.

Hypotheses / gap: Nondegenerate limiting law; positive normalizers; endpoint convergence alone is insufficient.

T012 Payoff versus indicator

P1 | source-check | TAIL | Printed pp. 18;225-227 | Source 2.2.23;11.2

Prove a weighted payoff integral is generally not threshold payoff times tail probability; characterize equality almost everywhere.

Hypotheses / gap: For an iff require nonnegative excess or monotonic payoff; cancellation invalidates unrestricted pointwise iff.

Chapter 3

T013 Overview and illustrative calculations

P3 | model-needed | APP | Printed pp. 25-67 | Source 3.1-3.13

Formalize only defined finite examples, tail ratios, and ruin models; link repeated claims to later results.

Hypotheses / gap: Qualitative philosophy and empirical claims are not Lean propositions until a model is supplied.

T014 Multiplicative survival and ruin

P1 | missing | RUIN | Printed pp. 53;63-66 | Source 3.6.1;3.11

Compute survival over n independent trials and its limit when each trial has positive ruin probability.

Hypotheses / gap: Independent trials or explicit conditional hazard lower bound; absorbing ruin state.

Chapter 4

T015 Gaussian scale mixture normalization

P1 | missing | MOM | Printed pp. 69-74 | Source 4.1

Build a probability mixture and prove its first four moments from conditional Gaussian moments.

Hypotheses / gap: Positive scales; probability mixing measure; integrability for each moment.

T016 Variance-preserving fattening

P1 | missing | MOM | Printed pp. 71-72 | Source 4.1.1

Show symmetric mixing of variances $\sigma^2(1-a)$ and $\sigma^2(1+a)$ keeps variance and raises fourth moment.

Hypotheses / gap: $0 \leq a < 1$; centered Gaussian components; nonzero σ for standardized kurtosis.

T017 Gamma mixture density

P3 | [missing](#) | [DENS](#) | Printed pp. 72-74 | [Source 4.1.2;4.7](#)

Derive the Bessel-K density for a Gaussian with gamma-distributed variance.

Hypotheses / gap: Positive gamma parameters; justified integral and transform; distinguish variance from precision.

T018 Precision mixture and Student law

P2 | [missing](#) | [DENS](#) | Printed pp. 74-75 | [Source 4.2](#)

Derive Student-type tails from gamma precision mixing; contrast lognormal and gamma variance mixtures.

Hypotheses / gap: Exact mixing parametrization; all moment finiteness claims must be checked separately.

T019 Crossovers of perturbed Gaussian densities

P1 | [missing](#) | [CALC](#) | Printed pp. 75-79 | [Source 4.3](#)

Locate the body/shoulder/tail sign changes through scale derivatives.

Hypotheses / gap: Centered Gaussian; positive scale; distinguish infinitesimal perturbation from a finite mixture.

T020 Moment and norm comparisons

P1 | [reuse](#) | [MOM](#) | Printed pp. 79-86 | [Source 4.4.1-4.4.4](#)

Apply Jensen, Holder, and L_p monotonicity to compare absolute mean deviation and standard deviation.

Hypotheses / gap: Probability measure; finite relevant moments; center the variable explicitly.

T021 Distribution-specific absolute moments

P1 | [missing](#) | [MOM](#) | Printed pp. 80-86 | [Source 4.4.2-4.4.4](#)

Compute Gaussian, Student, and Pareto absolute moments and MAD/STD ratios.

Hypotheses / gap: Correct scale convention; Pareto $\alpha > 2$ for STD; moment-order restrictions.

T022 Estimator efficiency comparison

P3 | [missing](#) | [EST](#) | Printed pp. 83 | [Source 4.4.3](#)

Derive asymptotic variances of MAD and standard-deviation estimators under specified laws.

Hypotheses / gap: Define estimator and efficiency criterion; finite moments; differentiate statistical functional.

T023 L_p -ball volumes

P2 | [source-check](#) | [GAMMA](#) | Printed pp. 88-89 | [Source 4.5;4.15](#)

Prove $\text{volume} = (2 \Gamma(1+1/p))^d / \Gamma(1+d/p)$, and limiting cube/cross-polytope cases.

Hypotheses / gap: $p > 0$; natural dimension; inspect the printed factor 4 before transcription.

Chapter 5**T024 One large jump and absent exponential moments**

P2 | [missing](#) | [SUBEXP](#) | Printed pp. 91-93 | [Source 5.1-5.2](#)

Prove fixed- n subexponential tail asymptotics and divergence of every positive exponential moment.

Hypotheses / gap: Nonnegative iid law; subexponential hypothesis; distinguish the converse and concentration-theory terminology.

T025 Exponential boundary

P1 | [source-check](#) | [SUBEXP](#) | Printed pp. 91-94 | [Source 5.0.1-5.0.3](#)

Show an exponential law fails the heavy-tail subexponential convolution test.

Hypotheses / gap: $\text{Rate} > 0$; its two-fold convolution tail ratio is $1 + \text{rate} \cdot x$, not 2.

T026 Regular-variation uniform convergence

P2 | [missing](#) | [RV](#) | [Printed pp. 95-98](#) | [Source 5.1](#)

Prove uniformity on compact positive scale intervals.

Hypotheses / gap: Positive measurable/Baire regular variation; not the ratio-only predicate without regularity.

T027 Potter bounds

P2 | [missing](#) | [RV](#) | [Printed pp. 95-98](#) | [Source 5.1](#)

Bound $L(y)/L(x)$ by a constant times $\max((y/x)^\epsilon, (y/x)^{-\epsilon})$ eventually.

Hypotheses / gap: Measurable slowly varying positive L ; $\epsilon > 0$; explicit threshold and constant.

T028 Tail moment threshold

P1 | [source-check](#) | [MOM](#) | [Printed pp. 95-98](#) | [Source 5.1](#)

Prove finite q th moment for $q < \alpha$ and infinite q th moment for $q > \alpha$; isolate $q = \alpha$.

Hypotheses / gap: $q > 0$; nonnegative law with regularly varying tail of index $-\alpha$; boundary requires an integral test on L .

T029 Tail of sums with unequal indices

P2 | [missing](#) | [SUBEXP](#) | [Printed pp. 98](#) | [Source 5.2.1](#)

Prove the heavier regularly varying tail dominates a sum under appropriate independence and support conditions.

Hypotheses / gap: Nonnegative independent variables; positive tails; account for cancellation for signed dependent laws.

T030 Product of exact Pareto laws

P1 | [missing](#) | [DENS](#) | [Printed pp. 99-100](#) | [Source 5.2.2](#)

Derive the product survival function for unequal indices and the logarithmic factor at equal indices.

Hypotheses / gap: Independent Pareto variables; positive scales and indices.

T031 Breiman product theorem

P2 | [source-check](#) | [RV](#) | [Printed pp. 100](#) | [Source Principle 5.2](#)

Prove $\text{tail}(XY) \sim E[Y^\alpha] \text{tail}(X)$ in its standard valid form.

Hypotheses / gap: $X \geq 0$ regularly varying; independent $Y \geq 0$; $E[Y^{\alpha+\epsilon}]$ finite; nonzero α -moment.

T032 Power transformation of a law

P1 | [partial](#) | [TAIL](#) | [Printed pp. 100](#) | [Source Property 5.2;5.8](#)

Bridge the checked analytic exponent α/p to the pushforward law of X^p .

Hypotheses / gap: $X \geq 0$; $p > 0$; measurability; monotonic inverse; do not apply to arbitrary negative p or signed X .

T033 Bell shape, interpolation, and log-Pareto

P2 | [missing](#) | [DENS](#) | [Printed pp. 101-103](#) | [Source 5.3-5.5](#)

Derive the stated Student/Pareto examples and exponential-of-Pareto survival and moments.

Hypotheses / gap: Exact distribution required; infinite moments use nonnegative integrals, not totalized real integrals.

T034 Stochastic-volatility interpretations

P3 | [model-needed](#) | [APP](#) | [Printed pp. 103-105](#) | [Source 5.6](#)

Separate exact mixture equivalences from finite-sample visual fits.

Hypotheses / gap: Specify mixing law and equality of laws; simulation fit is not proof.

Chapter 6

T035 Elliptical and multivariate Student laws

P2 | [missing](#) | [MULTI](#) | Printed pp. 108-114 | [Source 6.1-6.3](#)

Define characteristic/density forms, normalization, marginals, and linear projections.

Hypotheses / gap: Positive-definite shape matrix; distinction between scale matrix and covariance; dimension constraints.

T036 Uncorrelated versus independent

P1 | [missing](#) | [MULTI](#) | Printed pp. 113-114 | [Source 6.3.1](#)

Give a non-Gaussian elliptical example with zero covariance but dependent coordinates.

Hypotheses / gap: Finite second moments for covariance; exact joint law; independence needs a separate theorem.

T037 Student mutual information

P3 | [missing](#) | [ENT](#) | Printed pp. 114-115 | [Source 6.4](#)

Compute mutual information from joint and marginal densities and study its tail parameter.

Hypotheses / gap: Finite entropy/integrability; gamma and digamma identities; not Pearson correlation alone.

T038 Random matrices and undefined correlation

P3 | [model-needed](#) | [MULTI](#) | Printed pp. 115-117 | [Source 6.5-6.6](#)

State selected spectral/concentration claims and identify when covariance is undefined.

Hypotheses / gap: Explicit matrix ensemble and moment hypotheses; broad PCA failure slogans require a model.

T039 Regression with fat-tailed residuals

P3 | [model-needed](#) | [EST](#) | Printed pp. 117-119 | [Source 6.7](#)

Derive estimator laws or inconsistency/rate claims under a specified noise design.

Hypotheses / gap: Design matrix assumptions; independent residuals or explicit dependence; finite relevant moments.

Chapter A

T040 Mixture multimodality

P1 | [missing](#) | [DENS](#) | Printed pp. 122-124 | [Source A.1](#)

Determine modes and tail properties of a two-regime Gaussian mixture.

Hypotheses / gap: Weight in $[0,1]$; positive scales; separate multimodality from asymptotic heavy tails.

T041 Eventual transition or failure

P2 | [missing](#) | [RUIN](#) | Printed pp. 124-125 | [Source A.2](#)

Prove eventual hitting under stated transition probabilities.

Hypotheses / gap: Finite Markov chain or independent hazards; irreducibility/positive reachability hypotheses.

Chapter B

T042 Gaussian maximum entropy

P2 | [missing](#) | [ENT](#) | Printed pp. 128-129 | [Source B.1.1](#)

Show Gaussian uniquely maximizes differential entropy at fixed mean and variance.

Hypotheses / gap: Finite positive variance; admissible densities and entropy; prove global optimum via relative entropy.

T043 Pareto maximum entropy

P2 | [missing](#) | [ENT](#) | [Printed pp. 129](#) | [Source B.1.2](#)

Show Pareto maximizes entropy on $[L, \infty)$ with a fixed log moment.

Hypotheses / gap: $L > 0$; feasible finite log moment; normalization; integrability; equality up to almost everywhere.

T044 Uniform maximum entropy and probability integral transform

P1 | [missing](#) | [CDF](#) | [Printed pp. 129](#) | [Source B.1.3](#)

Prove uniform entropy bound; prove continuous CDF applied to its own random variable is uniform.

Hypotheses / gap: Finite interval; continuous CDF; generalized inverse for non-strictly increasing CDF.

T045 Tsallis entropy

P3 | [missing](#) | [ENT](#) | [Printed pp. 129-132](#) | [Source B.2](#)

Prove Shannon limit as q tends to 1 and derive admissible q -exponential optimizer.

Hypotheses / gap: Domination to exchange limit/integral; q -dependent normalization and moment domains.

Chapter 7

T046 Stable characteristic function specializations

P1 | [partial](#) | [CF](#) | [Printed pp. 139-140](#) | [Source 7.2.1](#)

Connect repaired expressions to actual Gaussian and Cauchy probability laws.

Hypotheses / gap: Scale convention: Gaussian variance = $2 \cdot \text{scale}^2$; Cauchy nonnegative scale; use existing Gaussian theorem.

T047 Stable sample averages

P2 | [partial](#) | [CF](#) | [Printed pp. 140](#) | [Source 7.2.2](#)

Derive the average scale $n^{1/(\alpha-1)}$, location rules, and finite-mean convergence.

Hypotheses / gap: Existence of stable laws; iid summands; $\alpha > 1$ for mean; $\alpha = 1$ skew location under rescaling needs logarithmic correction.

T048 Explicit finite-sum laws

P2 | [missing](#) | [DENS](#) | [Printed pp. 141-145](#) | [Source 7.3](#)

Compute uniform, exponential/gamma and selected Pareto convolutions; quantify errors in approximations.

Hypotheses / gap: iid laws; positive sample size; distinguish exact density, weak limit, and numerical plot.

T049 Cumulant scaling

P1 | [missing](#) | [MOM](#) | [Printed pp. 145-147](#) | [Source 7.4; Definition 7.1](#)

Show k th cumulant of normalized iid sum scales as $n^{1-k/2}$ when defined.

Hypotheses / gap: Moment/cumulant existence or local MGF assumptions; nonzero variance.

T050 Lyapunov implies Lindeberg

P2 | [missing](#) | [CLT](#) | [Printed pp. 147-148](#) | [Source 7.5](#)

Prove the implication and a CLT under the appropriate array assumptions.

Hypotheses / gap: Positive total variance; independent array; $\delta > 0$; uniform negligibility where required.

T051 LLN for powers and MS plots

P1 | [reuse](#) | [LLN](#) | [Printed pp. 148-150](#) | [Source 7.6](#)

Apply strong law to absolute powers; prove the maximum/sum criterion in its valid domain.

Hypotheses / gap: Finite p th absolute moment; nonnegative transformed variables; avoid inferring a theorem from a finite plot.

T052 Stable mean absolute deviation

P2 | [source-check](#) | [CF](#) | Printed pp. 151-152 | [Source 7.7;7.6-7.9](#)

Prove an absolute-moment formula using a convergent integral and fix Fourier/Hilbert constants.

Hypotheses / gap: $1 < \alpha \leq 2$; centered stable law; justified integral exchange; check Gaussian endpoint against $2 \cdot \text{scale} / \sqrt{\pi}$.

Chapter 8**T053 Kappa algebra and telescoping**

P1 | [source-check](#) | [KAPPA](#) | Printed pp. 156-158 | [Source Definition 8.1;8.1-8.2](#)

Prove the explicit log-ratio formula and the composition over intermediate sample sizes.

Hypotheses / gap: $n > n_0 \geq 1$; positive nonzero MAD ratios; sum from n_0 to $n-1$, avoiding $\log(0)$.

T054 Stable kappa and sample equivalence

P1 | [missing](#) | [KAPPA](#) | Printed pp. 158-160 | [Source 8.3](#)

Derive $\kappa = 2 - \alpha$ for finite-mean strictly stable cases and equal-MAD sample-size relations.

Hypotheses / gap: $1 < \alpha \leq 2$; correct centering and stable parameterization; scale positivity.

T055 Kappa bounds and non-universality

P1 | [source-check](#) | [KAPPA](#) | Printed pp. 160-162 | [Source 8.4](#)

State and prove bounds only for a specified distribution class.

Hypotheses / gap: Finite first moment alone does not imply $0 \leq \kappa \leq 1$; negative examples occur later in source.

T056 Cubic Student summed MAD

P2 | [missing](#) | [KAPPA](#) | Printed pp. 164-166 | [Source 8.6.1;Proposition 8.1](#)

Derive characteristic function, summed density, MAD, incomplete-gamma expression, and asymptotic κ .

Hypotheses / gap: iid Student with $\text{df}=3$ and stated scale; $n \geq 1$; all integral constants checked.

T057 Lognormal sum bounds

P2 | [missing](#) | [KAPPA](#) | Printed pp. 166-168 | [Source 8.6.2](#)

Prove low/high-volatility and sample-size bounds used for κ .

Hypotheses / gap: Independent lognormal variables; exact parameter convention; numerical approximations separately labeled.

T058 Exponential summed MAD

P1 | [missing](#) | [KAPPA](#) | Printed pp. 168-169 | [Source 8.6.3](#)

Derive gamma-sum MAD and explicit κ formula.

Hypotheses / gap: Positive rate; iid summands; $n \geq 2$ for nonzero log denominator.

T059 Negative kappa examples

P1 | [missing](#) | [KAPPA](#) | Printed pp. 169-170 | [Source 8.6.4](#)

Provide exact bounded/discrete counterexamples and identify failure of proposed universal bounds.

Hypotheses / gap: Exact distribution and MAD calculation; positive denominators.

Chapter 9

T060 Distribution of iid maxima

P1 | [missing](#) | [CDF](#) | [Printed pp. 171-173](#) | [Source 9.1](#)

Prove $\text{CDF}(\max_i X_i)(x) = F(x)^n$ and minima survival analog.

Hypotheses / gap: Independent measurable variables with common law; $n > 0$; atoms permitted.

T061 GEV families as probability measures

P1 | [partial](#) | [EVT](#) | [Printed pp. 172-173](#) | [Source 9.1](#)

Construct global Gumbel, Frechet and reverse-Weibull CDFs and prove validity.

Hypotheses / gap: Correct piecewise support; $\text{scale} > 0$; shape domains; right continuity and endpoint limits.

T062 Frechet and Gaussian maximum domains

P2 | [missing](#) | [EVT](#) | [Printed pp. 173-175](#) | [Source 9.1.1-9.1.2](#)

Prove regular-variation-to-Frechet convergence and Gaussian Gumbel normalizers.

Hypotheses / gap: Regularly varying tail; generalized inverse norming; Gaussian Mills asymptotics.

T063 Pickands-Balkema-de Haan

P2 | [missing](#) | [EVT](#) | [Printed pp. 176-177](#) | [Source 9.1.3;9.8-9.9](#)

Formalize threshold excess convergence to GPD and converse under correct hypotheses.

Hypotheses / gap: MDA hypothesis; moving endpoint; uniform metric; admissible shape-dependent support.

T064 Hidden moment law

P1 | [source-check](#) | [TAIL](#) | [Printed pp. 177-179](#) | [Source Proposition 9.1;9.12](#)

Derive distribution of tail moment above the random maximum in the finite-moment regime.

Hypotheses / gap: Exact Pareto or separately quantified asymptotic approximation; $p < \alpha$ for finite hidden pth moment; correct printed $p > \alpha$.

T065 Empirical CDF versus tail payoff

P2 | [missing](#) | [CLT](#) | [Printed pp. 179-181](#) | [Source 9.2.1;9.3](#)

Separate pointwise Bernoulli CLT, uniform empirical-process convergence, and unbounded-payoff estimation.

Hypotheses / gap: Continuity for probability integral transform; tightness for Donsker; uniform integrability for payoff limits.

Chapter C

T066 Log of a Pareto variable

P1 | [source-check](#) | [DENS](#) | [Printed pp. 183-185](#) | [Source Theorem 1](#)

Prove $\log(X/L)$ is exponential for exact Pareto X ; treat bilateral model separately.

Hypotheses / gap: $X \geq L > 0$ and $\alpha > 0$; full support alone does not force a Laplace law.

Chapter D

T067 Large deviation principle

P2 | [missing](#) | [CLT](#) | [Printed pp. 187-188](#) | [Source D](#)

State Cramer theorem and Chernoff rate bounds in the finite-MGF regime.

Hypotheses / gap: iid variables; log-MGF finite near zero; exponential tightness; no blanket extension to heavy tails.

Chapter E

T068 Pareto MLE and log transform

P1 | missing | EST | Printed pp. 189-191 | Source E;E.1

Derive $\alpha_{\text{hat}} = n / \sum \log(X_i/L)$, its inverse-gamma law and bias correction.

Hypotheses / gap: Known fixed L ; iid Pareto; $n > 1$ for estimator mean and $n > 2$ for variance.

T069 Tail-exponent sampling law

P2 | missing | EST | Printed pp. 191-192 | Source E.1

Prove finite-sample estimator moments, asymptotic variance and confidence constructions.

Hypotheses / gap: Exact Pareto model; distinguish arbitrary RV tails from parametric equality.

Chapter 10

T070 SP500 empirical claims

P3 | empirical | APP | Printed pp. 193-205 | Source 10.1-10.3

Reproduce diagnostics with data provenance and uncertainty; connect each diagnostic to a theorem.

Hypotheses / gap: Dataset and computation required; finite plots do not prove nonexistent moments.

T071 Aggregation and drawdown statistics

P2 | missing | MOM | Printed pp. 195-198 | Source 10.2.1-10.2.3

Prove model-specific kurtosis aggregation and drawdown/kappa formulas.

Hypotheses / gap: Independent increments where used; finite fourth moment for kurtosis; specify temporal dependence.

T072 Relative excess normalization

P1 | source-check | TAIL | Printed pp. 200 | Source Definition 10.1

Correct signs and distinguish $E[X \text{ given } X > K]/K$ from mean-excess/ K .

Hypotheses / gap: Pareto $\alpha > 1$ gives $\alpha/(\alpha-1)$; mean-excess ratio is $1/(\alpha-1)$.

T073 Record counts and asymmetric tails

P1 | missing | CDF | Printed pp. 201-205 | Source 10.2.7-10.2.8

Derive distribution-free record expectations for continuous iid samples and separate left/right tail diagnostics.

Hypotheses / gap: No ties or explicit tie rule; iid for record laws; data claims remain empirical.

Chapter F

T074 Risk estimator performance

P3 | model-needed | EST | Printed pp. 208-212 | Source F.1-F.2

Define loss and derive or simulate parametric/nonparametric estimator error under specified tails.

Hypotheses / gap: Specified model, sample size and loss; do not interpret finite simulation as universal dominance.

Chapter G

T075 Learning losses and angular calibration

P3 | model-needed | EST | Printed pp. 213-216 | Source G;G.0.1

Formalize moment/existence conditions for losses and proposed angle estimator.

Hypotheses / gap: Exact estimator; input normalization; integrability; predictive validity requires data.

Chapter 11

T076 Binary and unbounded payoff interfaces

P1 | missing | TAIL | Printed pp. 220-225 | Source Definitions 11.1-11.4

Define measurable indicator payoffs, unbounded exposures and characteristic-scale limits.

Hypotheses / gap: Measurability; support and integrability; source terminology differs from standard tail classifications.

T077 Payoff-to-binary tail ratios

P2 | missing | RV | Printed pp. 225-227 | Source Definition 11.5;Theorem 2

Derive I1/I2 from conditional means; prove Karamata ratio $\alpha/(\alpha-1)$ for regular variation.

Hypotheses / gap: $g(x)=x$ for displayed theorem; $\alpha>1$; positive survival; finite tail integral.

T078 Gaussian tail integral and Mills ratio

P1 | missing | TAIL | Printed pp. 226 | Source 11.6

Prove exact Gaussian tail first moment and asymptotic Mills ratio.

Hypotheses / gap: Positive scale; normalized density; correct quantile direction.

T079 Adjusted payoff-equivalent probability

P1 | source-check | TAIL | Printed pp. 228 | Source Definition 11.6

Use $p_{\text{star}}/p=\alpha/(\alpha-1)$ for exact Pareto and explain possible $p_{\text{star}}>1$.

Hypotheses / gap: $\alpha>1$; matching a monetary payoff is not necessarily a probability; printed denominator sign is wrong.

T080 Calibration versus payoff calibration

P2 | missing | EST | Printed pp. 230-231 | Source 11.3

Construct calibrated binary forecasts with different exposure losses.

Hypotheses / gap: Specify joint law of forecasts and outcomes; calibration condition and loss function.

T081 Bounded Brier score moments

P1 | missing | MOM | Printed pp. 231-238 | Source Theorem 3;11.15-11.16

Prove score in $[0,1]$ and all nonnegative moments finite; then a correctly normalized CLT.

Hypotheses / gap: Forecasts in $[0,1]$; $n>0$; no independence needed for boundedness; iid/variance for CLT.

T082 Score distribution transforms

P3 | missing | EST | Printed pp. 233-240 | Source 11.4.1;11.7

Derive binary tally and Brier score laws under the stated beta/independence model.

Hypotheses / gap: Independence of forecast errors and any outcomes as used; $\beta>0$; normalize special functions.

T083 Nonlinear loss moments

P1 | missing | TAIL | Printed pp. 235-238 | Source 11.5

Relate polynomial payoff growth to tail-index moment thresholds.

Hypotheses / gap: Measurable payoff; growth bounds; no unrestricted claim for all nonlinear losses.

Chapter 12

T084 Single forecasts and asymmetric decisions

P3 | model-needed | APP | Printed pp. 241-251 | Source 12.1-12.4

Define losses, Bayes decisions, threshold exceedance and ruin probability in explicit models.

Hypotheses / gap: Expected-loss existence; loss asymmetry; policy and empirical conclusions are not mathematical consequences alone.

Chapter 13

T085 Beta uncertainty in quantiles

P1 | missing | MOM | Printed pp. 253-257 | Source 13.1-13.3

For $P \sim \text{Beta}(a, b)$, $K = L \cdot P^{(-1/\alpha)}$, derive $E[K^q] = L^q \frac{B(a - q/\alpha, b)}{B(a, b)}$.

Hypotheses / gap: $L > 0$; $\alpha > 0$; $q \geq 0$; $a > q/\alpha$ for finiteness, otherwise divergent.

T086 Quantile sensitivity

P1 | missing | CALC | Printed pp. 257-259 | Source 13.3-13.5

Derive derivative/convexity and propagation of probability uncertainty into quantile risk.

Hypotheses / gap: $P \in (0, 1)$; $\alpha > 0$; use extended moments when divergent.

Chapter 14

T087 Gaussian digital payoff valuation

P3 | missing | SDE | Printed pp. 263-268 | Source 14.0.1; 14.1

Derive conditional normal exceedance probability and its time-to-maturity evolution.

Hypotheses / gap: Specified Brownian filtration; finite horizon; conditional expectations; no-arbitrage model explicit.

T088 Sigmoid of a martingale

P0 | source-check | SDE | Printed pp. 268 | Source Proposition 14.1

Replace the initial-value criterion by an Ito drift condition or conditional-expectation proof.

Hypotheses / gap: Starting at the sigmoid inflection point does not make a nondegenerate transformed process a martingale; constants also refute the literal only-if.

T089 Bounded dual diffusion

P3 | missing | SDE | Printed pp. 268-271 | Source 14.2-14.3

Solve the drift-adjusted process and prove transformed process is a martingale.

Hypotheses / gap: Specify SDE solution, adaptedness and integrability; local martingale versus true bounded martingale.

T090 Probability assessor and uniform forecast

P2 | missing | CDF | Printed pp. 271-273 | Source 14.4

Derive probability-integral-transform and forecasting implications without extra epistemic claims.

Hypotheses / gap: Continuous specified distribution; conditional law distinguished from unconditional law.

Chapter 15

T091 Population and sample Gini

P2 | [missing](#) | [GINI](#) | Printed pp. 277-282 | [Source 15.1-15.2](#)

Connect pairwise absolute differences, Lorenz/quantile integrals, order statistics and Gini mean deviation.

Hypotheses / gap: Nonnegative integrable X with $\text{mean} > 0$; atoms/ties handled; exact finite- n normalization.

T092 Order-statistic remainder

P2 | [missing](#) | [GINI](#) | Printed pp. 281-282 | [Source Lemma 15.1](#)

Prove weighted empirical-quantile remainder vanishes with the required normalization.

Hypotheses / gap: iid positive integrable samples; quantile conventions; uniform empirical-process control.

T093 Stable domain of attraction

P2 | [missing](#) | [CLT](#) | Printed pp. 282-283 | [Source 15.9;15.2.1](#)

Formalize two-sided tail balance and normalization for alpha-stable convergence.

Hypotheses / gap: $1 < \alpha < 2$; slowly varying norming; specified $S1/S0$ convention.

T094 Gini mean-deviation stable limit

P3 | [missing](#) | [GINI](#) | Printed pp. 283 | [Source Theorem 1](#)

Derive stable limit and consistency of GMD statistic including remainder.

Hypotheses / gap: Positive support; finite mean; RV tails; verify coupling of order statistics and transformed X_i .

T095 Gini ratio stable limit

P0 | [source-check](#) | [GINI](#) | Printed pp. 283-284;295 | [Source Theorem 2;15.42](#)

Derive joint numerator/denominator fluctuations before taking their ratio.

Hypotheses / gap: Random denominator fluctuations are of the same stable order; verify the scale using the centered variable $X_i \cdot (2F(X_i) - 1 - g)$.

T096 Parametric Gini MLE limit

P2 | [source-check](#) | [EST](#) | Printed pp. 284 | [Source Theorem 3](#)

Apply a valid MLE asymptotic normality theorem and delta method to the Gini functional.

Hypotheses / gap: Regular identifiable interior model; nonzero Fisher information; differentiable functional; exponential-family membership alone is insufficient.

T097 Pareto Gini estimator laws

P2 | [source-check](#) | [GINI](#) | Printed pp. 285-287 | [Source Corollaries 15.1-15.2](#)

Derive Pareto Gini $1/(2\alpha-1)$, MLE delta variance, and corrected stable asymptotics.

Hypotheses / gap: $\alpha > 1$; distinguish asymptotic approximation from exact finite- n equality in law.

T098 Finite-sample Gini correction

P3 | [missing](#) | [GINI](#) | Printed pp. 287-295 | [Source 15.5;appendix](#)

Derive mode/centering correction and prove its stated asymptotic or finite- n behavior.

Hypotheses / gap: Do not infer finite- n bias from a limit mode without a justified approximation.

Chapter 16

T099 Quantile-contribution estimator

P1 | [missing](#) | [GINI](#) | Printed pp. 299-302 | [Source 16.2;Proposition 16.1](#)

Define threshold/quantile shares and prove derivative sign for one added large observation.

Hypotheses / gap: Positive sums; frozen rank region; n and q rounding; threshold membership stable locally.

T100 Aggregation of concentration

P0 | [source-check](#) | GINI | Printed pp. 302-306 | [Source Theorem 4](#)

Prove the precise sample or expectation inequality, including any covariance terms.

Hypotheses / gap: Sample partition assumptions; do not factor $E[\text{weight} \cdot \text{share}]$ without justification; quantile ranks matter.

T101 Bias and consistency of threshold share

P0 | [source-check](#) | GINI | Printed pp. 305 | [Source Theorem 5](#)

Prove consistency from LLN; repair strict bias and monotonicity assertions separately.

Hypotheses / gap: Finite positive mean; nontrivial threshold; superadditivity of $n \cdot a_n$ alone does not imply a_n increasing.

T102 Mixtures and concentration

P3 | [missing](#) | GINI | Printed pp. 306-309 | [Source 16.4-16.6](#)

Derive concentration under random tail indices and effects of increasing sample totals.

Hypotheses / gap: Mixture support bounded away from nonintegrable regimes or explicit integrability; distinction between fixed and moving thresholds.

Chapter 17

T103 Dual bounded-to-unbounded transform

P1 | [missing](#) | CALC | Printed pp. 314-316 | [Source 17.2](#)

Prove $\phi(y) = L \cdot H \cdot \log((H-y)/(H-L))$ is an increasing bijection $[L, H)$ to $[L, \infty)$.

Hypotheses / gap: $0 < L < H$; endpoints handled; inverse $y = H - (H-L) \cdot \exp(-(x-L)/H)$.

T104 Shadow density and moments

P2 | [missing](#) | MOM | Printed pp. 316-319 | [Source 17.3](#)

Push GPD through the inverse transform and compute conditional mean and higher bounded moments.

Hypotheses / gap: Positive GPD scale; admissible support; H finite; special-function evaluation and normalization.

T105 Truncation versus transformation

P2 | [missing](#) | DENS | Printed pp. 319-321 | [Source 17.4-17.5](#)

Prove distinct resulting laws and compare moments under matching parameters.

Hypotheses / gap: Specify hard truncation, conditioning or smooth transformation; empirical fit is separate.

Chapter 18

T106 Conflict-data inference

P3 | [empirical](#) | APP | Printed pp. 323-342 | [Source 18.1-18.6](#)

Separate data cleaning, thresholds, tail fit and robustness from the reusable shadow/GPD theorems.

Hypotheses / gap: Actual data and provenance required; historical interpretation is outside Lean proof scope.

T107 Rescaling and conditional expectations

P2 | [missing](#) | TAIL | Printed pp. 328-329 | [Source 18.3.1-18.3.2](#)

Formalize deterministic rescaling and conditional tail estimators.

Hypotheses / gap: Positive scaling and defined reference population; justify conditional expectation and dependence.

T108 GPD inference and bootstrap

P3 | [model-needed](#) | [EST](#) | [Printed pp. 333-339](#) | [Source 18.4-18.5](#)

Specify and validate threshold estimator, QQ/mean-excess diagnostics and bootstrap model.

Hypotheses / gap: Asymptotic bootstrap validity requires hypotheses; resampling experiments are software tasks.

Chapter H

T109 War-probability discussion

P3 | [model-needed](#) | [APP](#) | [Printed pp. 343-345](#) | [Source H](#)

Map quantitative claims to the conflict model and record philosophical claims separately.

Hypotheses / gap: No probability of a future war follows from a severity-tail model without an arrival model.

Chapter 19

T110 Pandemic tail and shadow moments

P3 | [empirical](#) | [APP](#) | [Printed pp. 347-361](#) | [Source 19.1-19.4](#)

Reuse the dual-transform/GPD results; reproduce estimates and sensitivity to data errors.

Hypotheses / gap: Data and threshold assumptions; finite population cap does not by itself establish an unbounded model's mean.

Chapter 20

T111 Recursive mixtures

P1 | [missing](#) | [MOM](#) | [Printed pp. 368-371](#) | [Source 20.1.1-20.1.2](#)

Define finite binary scale tree and normalize its mixture law.

Hypotheses / gap: $0 \leq a_i < 1$; positive base scale; independent or explicitly weighted branches.

T112 Constant-error moment products

P1 | [missing](#) | [MOM](#) | [Printed pp. 372-373](#) | [Source 20.4](#)

Derive second and fourth moment factors $(1+a^2)^N$ and $(1+6a^2+a^4)^N$.

Hypotheses / gap: Centered Gaussian mixture; finite N; clarify centered/noncentral moments.

T113 Absolute-moment invariance

P1 | [missing](#) | [MOM](#) | [Printed pp. 372-373](#) | [Source 20.1](#)

Show centered Gaussian scale mixtures preserve first absolute moment when mean scale is fixed.

Hypotheses / gap: Center zero; nonnegative scales; finite mean scale; does not automatically extend to nonzero location.

T114 Exploding moments versus limiting law

P0 | [source-check](#) | [CLT](#) | [Printed pp. 373-377](#) | [Source 20.1.3;20.3](#)

Prove divergence of moment sequence; independently identify any weak limit under a valid scaling.

Hypotheses / gap: Diverging finite-N moments do not prove the weak limit has infinite moments or Pareto tails.

T115 Decaying-error infinite products

P2 | [missing](#) | [MOM](#) | [Printed pp. 375](#) | [Source 20.6-20.8](#)

Prove moment products converge under summable squared perturbations and verify q-Pochhammer identities.

Hypotheses / gap: $0 \leq \lambda < 1$; correct indices and powers; direct product proof before special-function simplification.

T116 Additive error recursion

P1 | [source-check](#) | MOM | Printed pp. 376 | [Source 20.2.2](#)

Derive moments of the specified alternative recursion from its exact definition.

Hypotheses / gap: Separate sigma from variance; recheck dimensional consistency of displayed formulas.

Chapter 21**T117 Random-index mixture foundations**

P1 | [missing](#) | MOM | Printed pp. 379-382 | [Source 21.1-21.4](#)

Define a measurable family of Pareto/RV laws and its mixture.

Hypotheses / gap: Fixed scale/support conventions; kernel measurability; moment integrability; alpha-dependence of L explicit.

T118 Moment convexity in Pareto exponent

P1 | [partial](#) | CALC | Printed pp. 381 | [Source Proposition 21.1](#)

Prove $m_p(\alpha) = \text{scale}^p \alpha / (\alpha - p)$ is convex on $\alpha > p$ and apply Jensen.

Hypotheses / gap: $p > 0$; common positive scale; random $\alpha > p$; integrable mixing or extended expectation.

T119 Mixed expected shortfall

P0 | [source-check](#) | TAIL | Printed pp. 381-383 | [Source Proposition 21.2; 21.5-21.8](#)

Derive posterior tail-weighted mixing for $E[X \text{ given } X > K]$ and formulate a nonvacuous comparison.

Hypotheses / gap: Positive tails and finite moments; compare normalized shortfall or finite K; unscaled infinite limits give no quantitative inequality.

T120 Density and boundary normalization

P0 | [source-check](#) | RV | Printed pp. 381-382 | [Source 21.5; Karamata representation](#)

Differentiate $S(x) = L(x) * x^{-\alpha}$, enforce $S(x_0) = 1$ and integrate density.

Hypotheses / gap: Absolute continuity; nonnegative density; boundary terms; printed derivative-based normalizer must be rederived.

T121 Sums and stochastic indices

P2 | [missing](#) | SUBEXP | Printed pp. 383-384 | [Source 21.3](#)

Combine sum tails and mixture tails without interchanging limits illegitimately.

Hypotheses / gap: Independence; domination/uniform regular variation; essential infimum of mixing indices can govern tails.

T122 Asymmetric stable mixture

P2 | [missing](#) | CF | Printed pp. 384-385 | [Source 21.4](#)

Characterize which skew/mean properties survive mixing and which stability properties are lost.

Hypotheses / gap: Stable law existence; finite moments where used; a mixture of stable laws is not automatically stable.

T123 Shifted-lognormal index mean

P0 | [source-check](#) | MOM | Printed pp. 385 | [Source Proposition 21.3; 21.11-21.13](#)

Use $\text{scale} * (1 + E[1/(b-1+Z)])$; simplify only at $b=1$; audit the series exchange.

Hypotheses / gap: Z lognormal with stated mean/log-variance; $b \geq 1$; the displayed arbitrary-b closed form is false.

T124 Shifted-gamma index mean

P1 | partial | GAMMA | Printed pp. 386 | Source Proposition 21.4;21.14-21.16

Prove inverse gamma moment and corrected excess scale $s^2/(m(m^2-s^2))$.

Hypotheses / gap: $m=\alpha-1>0$; $s>0$; $m^2>s^2$ for finite mean; algebraic excess already checked.

T125 Mixed bounded power law

P2 | missing | MOM | Printed pp. 386-387 | Source 21.7

Compose index mixing with bounded dual transform and prove finite moments.

Hypotheses / gap: Finite H; common support; kernel and transformation measurable; no unbounded-mean claim after capping.

Chapter 22**T126 Finite-sample p-value density**

P0 | source-check | EST | Printed pp. 391-393 | Source Proposition 22.1;22.1

Derive pushforward density under the exact sampling model and normalize it.

Hypotheses / gap: Specify t statistic law under alternative; central shifted-t need not equal a noncentral t law; p in (0,1).

T127 Gaussian-limit p-value law

P2 | missing | EST | Printed pp. 392-394 | Source Proposition 22.2;22.2

Derive inverse-normal change of variables and convergence of p-value law.

Hypotheses / gap: Sample-statistic convergence; continuity; endpoint behavior; median parameter in (0,1).

T128 Minimum p-value law and expectation

P1 | source-check | CDF | Printed pp. 395 | Source Proposition 22.3;22.5

Prove $\text{survival}(\min)=\text{survival}^m$ and $E[\min]=\int_0^1 \text{survival}^m$.

Hypotheses / gap: Independent identical p-values; $m \geq 1$; correct negative displayed expectation formula.

T129 Power-of-test distribution

P3 | source-check | EST | Printed pp. 396 | Source Proposition 22.4;22.6-22.7

Specify estimator and derive its pushforward law, including endpoint limits.

Hypotheses / gap: Exact test statistic and alternative; distinguish estimated power from a fixed parameter.

Chapter I**T130 Loss-aversion examples**

P3 | model-needed | APP | Printed pp. 399-403 | Source I.1

Calculate expected loss or utility under an explicit multiplicative/heavy-tail process.

Hypotheses / gap: Specified utility and integrability; behavioral interpretation requires independent empirical evidence.

Chapter 23**T131 Survival, hazard and residual lifetime**

P1 | missing | RUIN | Printed pp. 405-409 | Source 23.1-23.2

Define hazard and mean residual life; prove Pareto residual life scales with age.

Hypotheses / gap: Positive survival; differentiability for hazard; $\alpha>1$ for finite remaining mean.

T132 Brownian absorbing barrier

P3 | [missing](#) | [SDE](#) | [Printed pp. 409-411](#) | [Source 23.2.1](#)

Derive first-passage distribution using reflection and study survival asymptotics.

Hypotheses / gap: Brownian motion with specified variance; positive initial distance; stopping-time measurability.

T133 Drifted barrier and Lindy interpretation

P3 | [missing](#) | [SDE](#) | [Printed pp. 411-413](#) | [Source 23.2.2-23.3](#)

Derive drifted hitting-time law and conditions for finite lifetime/aging.

Hypotheses / gap: Correct drift sign; Girsanov or direct PDE hypotheses; do not equate median and infinite mean.

Chapter 24

T134 Hedging-error decomposition

P3 | [model-needed](#) | [FIN](#) | [Printed pp. 417-420](#) | [Source 24.1](#)

Specify discrete hedge error and conditions under which quadratic approximations are valid.

Hypotheses / gap: Self-financing strategy, payoff regularity, integrability and jump process explicit.

Chapter 25

T135 Call-price curve determines a marginal law

P2 | [source-check](#) | [FIN](#) | [Printed pp. 423-426](#) | [Source Theorem 6;Lemma 25.1](#)

Reconstruct a terminal probability measure from a full arbitrage-free call curve.

Hypotheses / gap: All strikes, convexity, monotonicity, endpoint slopes and finite first moment; a single forward price is insufficient.

T136 Put/call law uniqueness

P2 | [missing](#) | [FIN](#) | [Printed pp. 426](#) | [Source Lemma 25.2](#)

Use put-call parity and full strike families to prove equality of induced terminal laws.

Hypotheses / gap: Correct one-sided derivatives at atoms; normalized laws and generating half-lines.

T137 Forward mean is not measure uniqueness

P0 | [source-check](#) | [FIN](#) | [Printed pp. 426](#) | [Source Lemma 25.3;25.13](#)

Replace invalid Radon-Nikodym conclusion with equality of first moments or full call-curve uniqueness.

Hypotheses / gap: Equal means alone permit distinct laws; payoff difference is $x-K$; pricing assumptions must be explicit.

T138 Physical versus pricing measure

P3 | [model-needed](#) | [FIN](#) | [Printed pp. 427](#) | [Source 25.3](#)

Formalize exactly which law prices which contracts and numeraire.

Hypotheses / gap: No automatic identification of physical and pricing laws; terminal marginal versus path law distinction.

Chapter 26

T139 Static payoff formulas and hedging

P2 | [missing](#) | [FIN](#) | [Printed pp. 434-439](#) | [Source 26.4](#)

Derive Bachelier/Black-Scholes payoff integrals under specified terminal laws.

Hypotheses / gap: Normal/lognormal parameters; nonnegative maturity/volatility; edge cases; numerical evaluation separate.

T140 Limits of dynamic replication

P3 | [model-needed](#) | [FIN](#) | Printed pp. 439-442 | [Source 26.6](#)

Formulate one precise impossibility or approximation result in a specified jump/heavy-tail model.

Hypotheses / gap: Trading strategies and admissibility defined; no universal impossibility inferred from heavy tails alone.

Chapter 27**T141 Pareto call-price tail**

P1 | [source-check](#) | [FIN](#) | Printed pp. 446-448 | [Source 27.1-27.3;Result 1](#)

Prove $C(K) = \int_K^\infty S(x) dx$ and exact power pricing for an exact Pareto tail.

Hypotheses / gap: $\alpha > 1$; finite first moment; fixed measure; RV only yields asymptotics unless L is exactly constant.

T142 Log return transforms

P1 | [missing](#) | [RV](#) | Printed pp. 448 | [Source Theorem 7](#)

Prove $\log(S/S_0)$ has exponentially decaying tail for Pareto S and fails a finite RV power index.

Hypotheses / gap: $S_0 > 0$; $\alpha > 0$; regular variation and suitable asymptotic transfer; repair source proof expression.

T143 Shifted Pareto relative calls

P1 | [missing](#) | [FIN](#) | Printed pp. 448-449 | [Source 27.4-27.6;Result 2](#)

Derive relative call ratios for arithmetic-return Pareto tails.

Hypotheses / gap: $K > S_0$; common exact tail region; $\alpha > 1$; anchor call price positive.

T144 Put-price power law

P0 | [source-check](#) | [FIN](#) | Printed pp. 449-451 | [Source 27.7;Result 3](#)

Derive left-tail/limited-liability put formulas with real-valued bases.

Hypotheses / gap: Specify support and truncation; negative bases with noninteger powers in displayed formula need correction.

T145 Splice no-arbitrage boundaries

P2 | [missing](#) | [FIN](#) | Printed pp. 451 | [Source 27.8-27.10](#)

Prove convexity and slope conditions at a splice of central and tail call-price curves.

Hypotheses / gap: Positive strike/maturity; correct derivative orientation; local inequalities versus global arbitrage freedom.

Chapter 28**T146 Second versus fourth moment**

P1 | [reuse](#) | [MOM](#) | Printed pp. 453-454 | [Source 28.1](#)

Prove variance estimator behavior requires more than existence of variance.

Hypotheses / gap: Finite fourth moment for variance of squared observations; specify estimator and LLN notion.

T147 Jensen, insurance and numeraire

P2 | [missing](#) | [FIN](#) | Printed pp. 454-456 | [Source 28.2-28.4](#)

Formalize payoff convexity effects and numeraire-consistent expectation comparisons.

Hypotheses / gap: Probability measure, payoff and units fixed; financing assumptions explicit.

T148 Tail betting examples

P3 | [model-needed](#) | [FIN](#) | Printed pp. 456-457 | [Source 28.5](#)

Compute exact expected payoffs in the specified betting model.

Hypotheses / gap: Integrability and position sizing; selected numerical examples are not investment guarantees.

Chapter 29**T149 Dependence beyond correlation**

P2 | [missing](#) | [MULTI](#) | Printed pp. 459-467 | [Source 29](#)

Construct laws with matching marginals/correlation but different tail dependence and portfolio loss.

Hypotheses / gap: Finite variances where correlation used; copula or joint law explicit; empirical portfolio claims separate.

T150 Portfolio expected shortfall

P3 | [model-needed](#) | [FIN](#) | Printed pp. 459-467 | [Source 29](#)

Derive risk bounds or optimization results under an explicit joint law and constraints.

Hypotheses / gap: Integrability; dependence; leverage and admissible set; no blanket theorem that all correlation use fails.

Chapter 30**T151 Tail constraint feasibility**

P1 | [missing](#) | [ENT](#) | Printed pp. 469-474 | [Source 30.1-30.2](#)

Characterize admissible distributions with mass epsilon below K and conditional mean ν_{minus} .

Hypotheses / gap: $0 < \epsilon < 1$; $\nu_{\text{minus}} < K$; finite conditional moments; total mean relation.

T152 Gaussian tail constraint solution

P2 | [missing](#) | [ENT](#) | Printed pp. 474;481 | [Source Proposition 30.1;30.2-30.3](#)

Derive the two linear equations, solve for mean/scale, and prove sign/limit properties of $B(\epsilon)$.

Hypotheses / gap: Gaussian quantile below zero; K and ν_{minus} conventions; positive resulting scale; Mills ratio bounds.

T153 Two-normal small-variance limit

P2 | [missing](#) | [CLT](#) | Printed pp. 475 | [Source 30.3.1](#)

Prove mixture constraints are approached as component variances vanish.

Hypotheses / gap: Weak convergence to two point masses; thresholds away from atom boundary; matching means.

T154 Entropy optimizer existence

P2 | [source-check](#) | [ENT](#) | Printed pp. 476-477 | [Source 30.4](#)

Prove entropy unbounded with tail-only constraints; give conditions for existence and uniqueness with extra constraints.

Hypotheses / gap: Nonempty feasible set alone does not guarantee an optimizer; entropy defined and finite on comparison class.

T155 Piecewise-exponential maximum entropy

P2 | [missing](#) | [ENT](#) | Printed pp. 477 | [Source 30.4.1](#)

Prove normalization, all three constraints, and global maximum for the candidate density.

Hypotheses / gap: $\nu_{\text{minus}} < K < \nu_{\text{plus}}$; ϵ in $(0,1)$; equality almost everywhere; Gibbs inequality.

T156 Absolute-moment entropy optimizer

P0 | [source-check](#) | ENT | Printed pp. 478 | [Source 30.4.2](#)

Normalize truncated Laplace part and determine λ from the absolute moment.

Hypotheses / gap: When $K < 0$, left contribution is $-\epsilon \nu_{\min}$, not $\epsilon \nu_{\min}$; existence/uniqueness of λ .

T157 Power-tail maximum entropy

P2 | [missing](#) | ENT | Printed pp. 479 | [Source 30.4.3](#)

Derive normalizer, logarithmic moment equation and uniqueness of α .

Hypotheses / gap: $K < 0$ for displayed normalizer; $\alpha > 0$; feasible log constraint; genuine global optimum.

T158 Multi-period averages and stable/Gaussian limits

P2 | [missing](#) | CLT | Printed pp. 480-481 | [Source 30.1;30.4.4](#)

Derive law of averages and correctly normalized fluctuation limits for each candidate law.

Hypotheses / gap: iid periods; finite first moment for LLN; finite variance or domain-of-attraction condition for Gaussian normalization.

Acceptance record and provenance

A backlog family becomes completed only after the formal proposition matches a corrected mathematical statement, required assumptions are explicit, its code compiles in a pinned project, the axiom closure is checked, and source differences are documented. Definitions and conditional bridges should retain their scope labels until the missing existence and model-identification theorems are supplied.

Exact source input

arXiv:2001.10488v4, third edition, 17 September 2025

Source PDF SHA-256:

758e18b7337840104db93296144d67cf
5514bf2de128fe557dc84bc32a0ad567

The original proof ZIP hash is verified and its untouched bytes are included under evidence/source. The source book and dependency binaries are not included. The archive has SHA256SUMS for its contents; verification.json separately hashes every Lean file.

Primary references

Versioned book: [arXiv:2001.10488v4](#)

[Pinned Mathlib source tree](#)

[Characteristic functions in the pinned source](#)

[Pareto distribution in the pinned source](#)

Prior evidence: [Taleb_Fat_Tails_Lean_Audit.pdf](#) and [Taleb_Lean_Checked_Repairs.zip](#), recovered from the previous completed audit. The earlier implementation is acknowledged as the foundation of this expanded handoff.

Verification scope: compilation validates the Lean propositions under their definitions and standard foundational axioms. It does not validate empirical conclusions, guarantee current upstream novelty, certify arbitrary numerical implementations, or establish a theorem that is still listed as a premise.